

Python for AI & Machine Learning

Class 26 – Advanced Data Visualization with Matplotlib & Seaborn

Course: Python for AI & Machine Learning

Module: Data Visualization using Python

Duration: 2 Hours (Theory + Practical)

Learning Objectives

By the end of this session, students will be able to:

- Understand the importance of advanced data visualization.
 - Customize Matplotlib figures.
 - Create multiple plots using subplots.
 - Learn the basics of Seaborn.
 - Create Box Plots, Violin Plots, Heatmaps, Pair Plots, and Count Plots.
 - Understand correlation analysis.
 - Apply visualization techniques in AI and Machine Learning projects.
-

Revision

In the previous class, we learned:

- Introduction to Matplotlib
- Line Charts
- Bar Charts
- Pie Charts
- Histograms
- Scatter Plots
- Titles, Labels, Legends, and Grid
- Saving Charts

Today, we will learn **Advanced Data Visualization** using **Matplotlib** and **Seaborn**.

Introduction to Seaborn

Seaborn is a Python data visualization library built on top of Matplotlib.

It provides:

- Attractive default themes
- Statistical visualizations
- Better color palettes
- Easy integration with Pandas DataFrames

It is widely used in:

- Data Science
 - Artificial Intelligence
 - Machine Learning
 - Business Analytics
-

Installing Seaborn

Install Seaborn using pip:

```
pip install seaborn
```

Verify installation:

```
pip show seaborn
```

Importing Libraries

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
```

Setting Seaborn Theme

```
import seaborn as sns

sns.set_theme(style="darkgrid")
```

Available themes:

- darkgrid

- whitegrid
 - dark
 - white
 - ticks
-

Figure Size

Change the size of a graph.

```
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))

plt.plot([10,20,30],[5,10,15])

plt.show()
```

Subplots

Subplots allow multiple charts in a single figure.

```
import matplotlib.pyplot as plt

plt.subplot(1,2,1)
plt.plot([1,2,3],[4,5,6])
plt.title("Chart 1")

plt.subplot(1,2,2)
plt.bar(["A", "B", "C"], [10,15,8])
plt.title("Chart 2")

plt.tight_layout()

plt.show()
```

Box Plot

A Box Plot shows:

- Minimum
- Maximum

- Median
- Quartiles
- Outliers

```
import seaborn as sns
import matplotlib.pyplot as plt

marks = [65,70,75,80,85,90,95,100]

sns.boxplot(y=marks)

plt.show()
```

Applications:

- Detecting outliers
- Data distribution analysis
- Data preprocessing

Violin Plot

A Violin Plot combines a Box Plot with a distribution curve.

```
import seaborn as sns
import matplotlib.pyplot as plt

marks = [60,70,75,80,85,90,95]

sns.violinplot(y=marks)

plt.show()
```

Count Plot

Displays the frequency of categories.

```
import seaborn as sns
import pandas as pd

students = pd.DataFrame({
    "Course": ["Python", "AI", "Python", "ML", "AI", "Python"]
})

sns.countplot(data=students, x="Course")
```

```
plt.show()
```

Heatmap

A Heatmap displays values using colors.

```
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt

data = pd.DataFrame(
    [
        [10,20,30],
        [40,50,60],
        [70,80,90]
    ]
)

sns.heatmap(data,annot=True)

plt.show()
```

Applications:

- Correlation analysis
- Feature importance
- Confusion matrices

Correlation Matrix

Shows relationships between numerical variables.

```
import pandas as pd

student = {
    "Math": [80,90,85,95],
    "Science": [75,88,82,94],
    "English": [70,92,80,90]
}

df = pd.DataFrame(student)
```

```
print(df.corr())
```

Visualize it using a Heatmap:

```
sns.heatmap(df.corr(),annot=True,cmap="coolwarm")

plt.show()
```

Pair Plot

Shows pairwise relationships between columns.

```
import seaborn as sns

iris = sns.load_dataset("iris")

sns.pairplot(iris)

plt.show()
```

Applications:

- Feature comparison
- Data exploration
- Machine Learning preprocessing

Line Plot using Seaborn

```
import seaborn as sns
import pandas as pd

sales = pd.DataFrame({
    "Month": ["Jan", "Feb", "Mar", "Apr"],
    "Revenue": [25000, 30000, 45000, 50000]
})

sns.lineplot(data=sales, x="Month", y="Revenue")

plt.show()
```

Bar Plot using Seaborn

```
import seaborn as sns
import pandas as pd

students = pd.DataFrame({
    "Course": ["Python", "AI", "ML"],
    "Students": [45, 35, 25]
})

sns.barplot(data=students, x="Course", y="Students")

plt.show()
```

AI Example 1 – Student Marks Distribution

```
import seaborn as sns
import pandas as pd

df = pd.DataFrame({
    "Marks": [65, 75, 82, 90, 95, 88, 76]
})

sns.histplot(df["Marks"])

plt.show()
```

AI Example 2 – Sales Correlation

```
import seaborn as sns
import pandas as pd

sales = pd.DataFrame({
    "Sales": [20, 25, 30, 35],
    "Profit": [5, 7, 10, 12]
})

sns.heatmap(sales.corr(), annot=True)

plt.show()
```

AI Example 3 – Employee Salary Analysis

```
import seaborn as sns
import pandas as pd

employees = pd.DataFrame({
    "Department": ["HR", "IT", "HR", "Sales", "IT"]
})

sns.countplot(data=employees, x="Department")

plt.show()
```

AI Example 4 – Model Accuracy

```
import seaborn as sns
import pandas as pd

accuracy = pd.DataFrame({
    "Epoch": [1, 2, 3, 4, 5],
    "Accuracy": [65, 72, 80, 88, 95]
})

sns.lineplot(data=accuracy, x="Epoch", y="Accuracy")

plt.show()
```

Advantages of Seaborn

- Beautiful default themes.
- Less code than Matplotlib.
- Easy statistical plotting.
- Works directly with Pandas DataFrames.
- Better color management.
- Excellent for Machine Learning visualization.

Matplotlib vs Seaborn

Matplotlib	Seaborn
Low-level plotting library	High-level visualization library

Matplotlib	Seaborn
Requires more code	Requires less code
Highly customizable	Attractive default styles
General-purpose graphs	Statistical graphics
Foundation library	Built on top of Matplotlib

Best Practices

- Choose the appropriate chart for your data.
 - Use readable titles and labels.
 - Avoid overcrowding charts.
 - Use Heatmaps to analyze correlations.
 - Use Pair Plots for feature analysis.
 - Use consistent colors across reports.
-

Practical Exercise 1

Create a Box Plot showing student marks.

Practical Exercise 2

Create a Count Plot showing student enrollment in different courses.

Practical Exercise 3

Create a Heatmap showing subject correlations.

Practical Exercise 4

Create a Pair Plot using the Iris dataset.

Practical Exercise 5

Create a dashboard with:

- Line Chart
- Bar Chart
- Heatmap

using Subplots.

Assignment

AI Student Analytics Dashboard

Create a dashboard that:

- Reads student data from a CSV file.
- Displays:
 - Line Chart
 - Bar Chart
 - Box Plot
 - Heatmap
 - Count Plot
- Displays a Correlation Matrix.
- Uses Seaborn themes.
- Saves the dashboard as **student_dashboard.png**.

Bonus Challenge:

Use the Iris dataset to create a Pair Plot and identify relationships between different flower measurements.

Interview Questions

1. What is Seaborn?
 2. How is Seaborn different from Matplotlib?
 3. What is a Box Plot?
 4. What is a Violin Plot?
 5. What is a Heatmap?
 6. What is a Pair Plot?
 7. What is a Count Plot?
 8. What is a Correlation Matrix?
 9. Why are Heatmaps useful in Machine Learning?
 10. What are the advantages of Seaborn?
-

Summary

Today we learned:

- Introduction to Seaborn
- Installing and Importing Seaborn
- Themes
- Figure Size
- Subplots
- Box Plot
- Violin Plot
- Count Plot
- Heatmap
- Correlation Matrix
- Pair Plot
- Seaborn Line and Bar Charts
- AI & Machine Learning Applications

Advanced visualization techniques help Data Scientists explore data, detect patterns, identify outliers, and understand relationships between features before building Machine Learning models.

Next Class Preview

Class 27 – Introduction to Machine Learning with Scikit-learn

Topics to be covered:

- What is Machine Learning?
- Types of Machine Learning
- Installing Scikit-learn
- Machine Learning Workflow
- Features and Labels
- Training and Testing Data
- First Machine Learning Model
- Model Evaluation
- Real-world AI Applications